Activity 1

17 (i) I Recommend Contrast Stretching to improve the visibility of darker tissues in the X-Ray image.

(ii) Contrast Stretching increases the range of gray levels in the image. It improves the contrast between darker tissues and brighter bones, making the details easier to identify.

(iii) The image will have better contrast. Darker tissues will become easier to distiguish, and the bones and surrounding tissues will be more clearly visible

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2) (i) I recommend Log Transformation to improve the visibility of the dark CCTV image

(ii) Log Transformation expands the low-intensity pixel values and compresses the high-intensity values. Since most pixel values are between 0

and so, it brightens the dark regions effectively. Negative transformation reverses the brightness values but does not specifically enhance dark intensity differences in the same way.

(iii) The enhanced image will appear brighter. Objects such as vehicles, people, and other details in the dark parking lot will become more visible.

3)

(i) I would apply contrast stretching to improve the contrast between fingerprint ridges and valleys.

(ii) Contrast Stretching expands the gray-level range, making the ridges and valleys easier to distigwish. This improves fingerprint detail visibility and supports accurate regeognition.

(iii) Log transformation would brighten the darker regions and compress the brightness

regions. However, it might not improve the contrast between ridges and valleys as effectively as contrast stretching.